



# Dark Matter and Dark Energy

## Common Misconceptions

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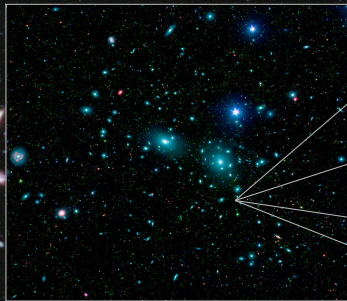
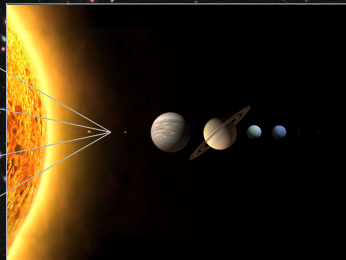
# Overview

Our Picture of the Universe

Dark Matter

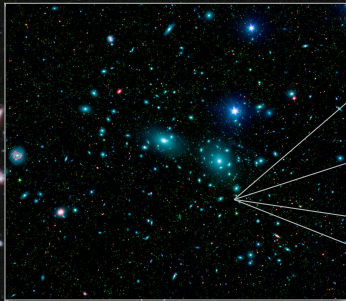
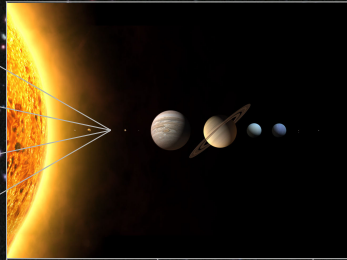
Dark Energy

# Our Picture of the Universe





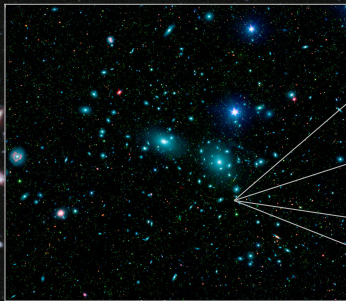
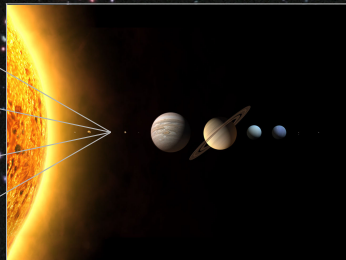
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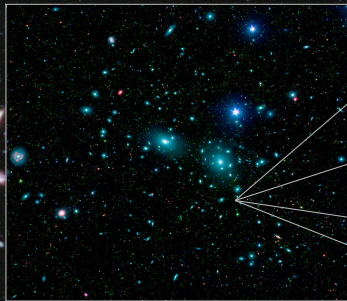
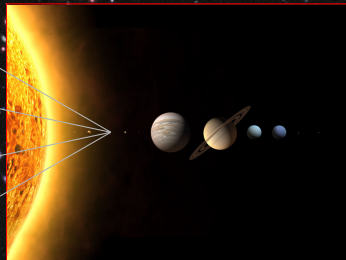
Earth



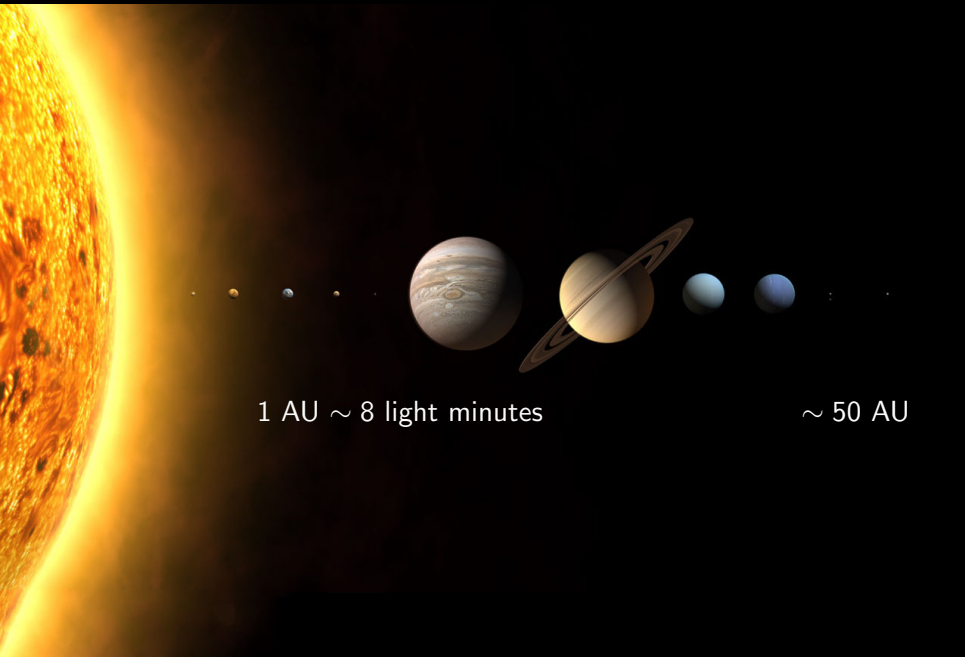
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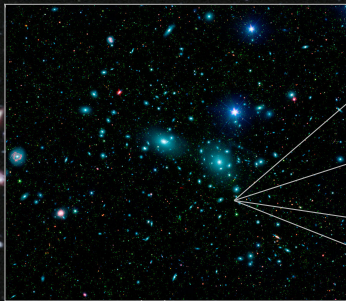
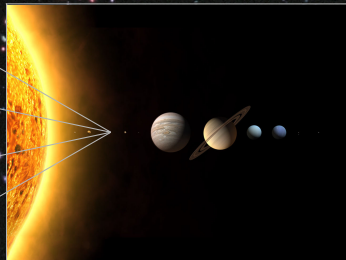


# The Solar System

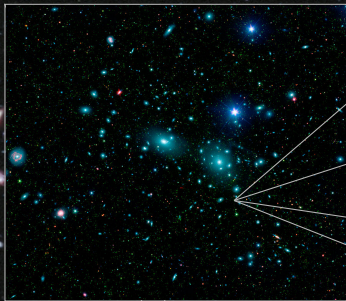
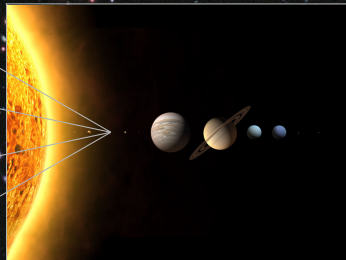




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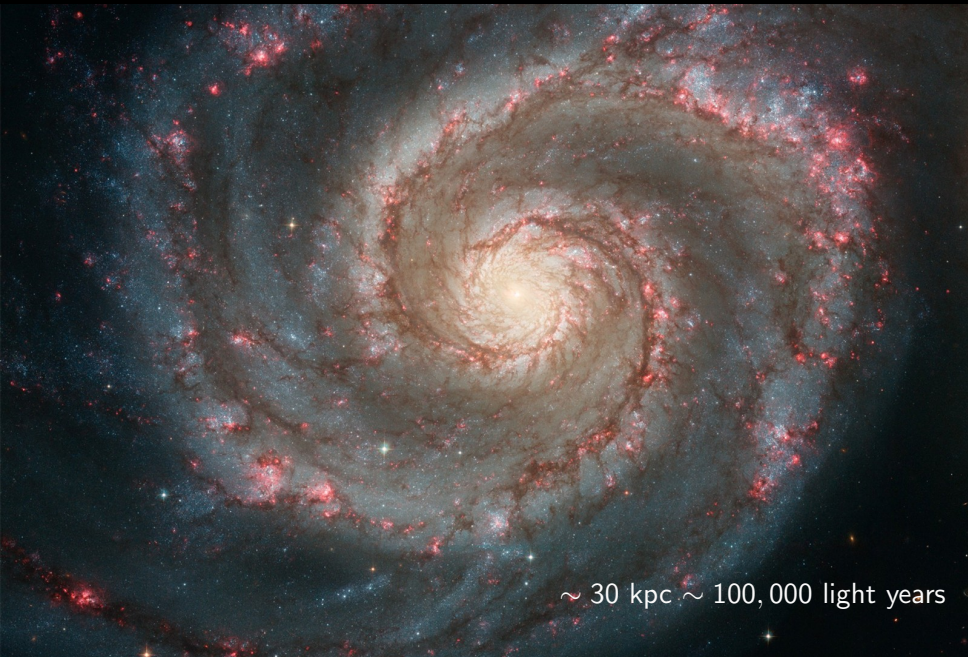


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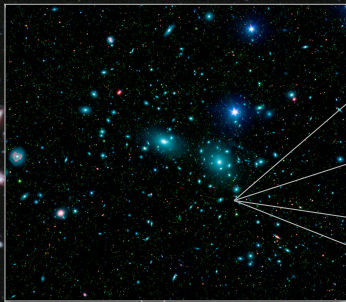
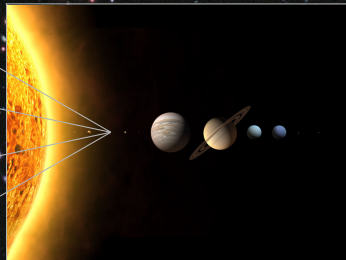


# Galaxies

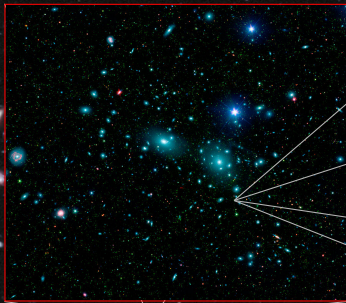
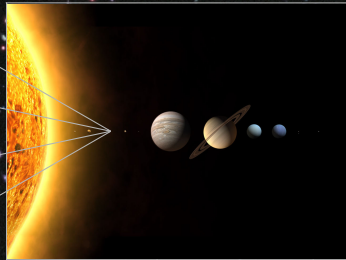


~ 30 kpc ~ 100,000 light years

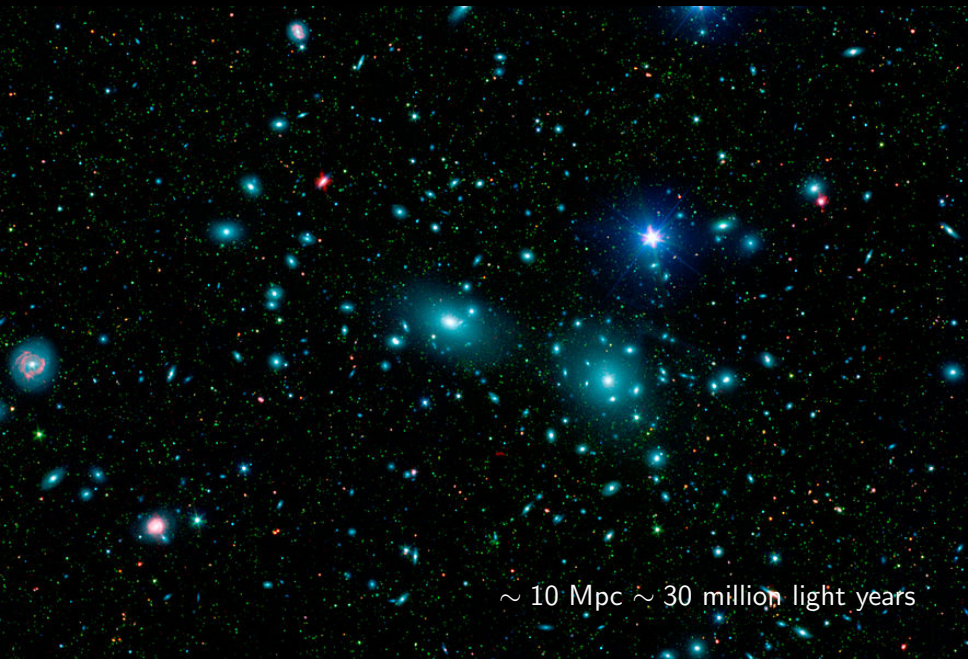
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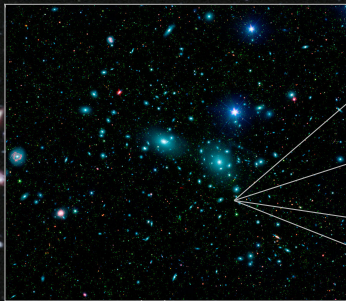
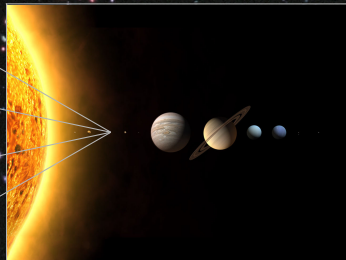
# Galaxy Clusters



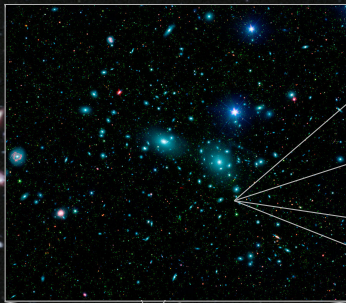
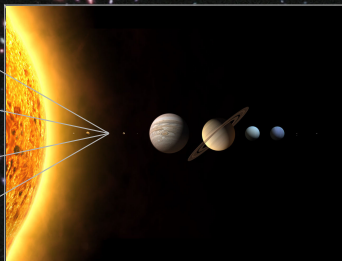
$\sim 10$  Mpc  $\sim 30$  million light years



# Our Picture of the Universe



# Our Picture of the Universe





# Galaxy Clusters

Hubble extreme deep field





# Dark Matter

What's missing?



# Dark Matter

What's missing?

- ▶ Galaxy cluster: gravitationally bound.



# Dark Matter

What's missing?

- ▶ Galaxy cluster: gravitationally bound.
- ▶ Count the matter we can see.





# Dark Matter

What's missing?

- ▶ Galaxy cluster: gravitationally bound.
- ▶ Count the matter we can see.
- ▶ There is not enough matter.



# What's missing

## The Problem



# What's missing

## The Problem

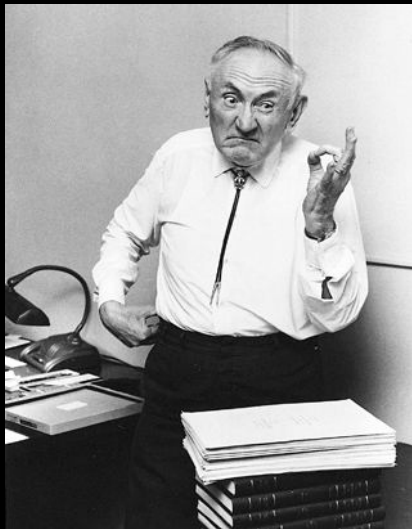


► Not enough stuff here either

# Dark Matter

## The Problem

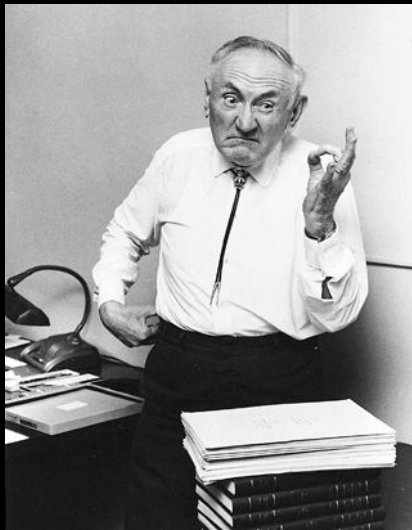
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# Dark Matter

## The Problem

- ▶ Observed by Fritz Zwicky in 1933.
- ▶ There appears to be a large amount of matter which we cannot see.

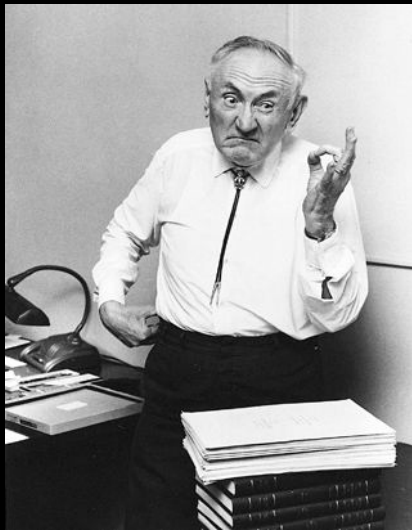




# Dark Matter

## The Problem

- ▶ Observed by Fritz Zwicky in 1933.
- ▶ There appears to be a large amount of matter which we cannot see.
- ▶ We inferred its existence from its *gravitational presence*



# Dark Matter as a prediction

In general, if you encounter a discrepancy between theory and observation, there are two options:

# Dark Matter as a prediction

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- ▶ Modify the theory

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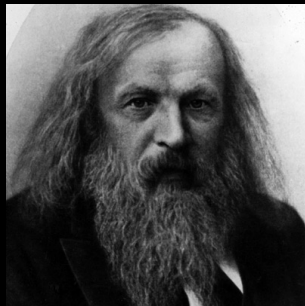
In general, if you encounter a discrepancy between theory and observation, there are two options:

- ▶ Modify the theory
- ▶ Predict a new object

# Example 1.

## The Periodic Table

- ▶ In 1865, Mendeleev proposed that elements be ordered into the periodic table.

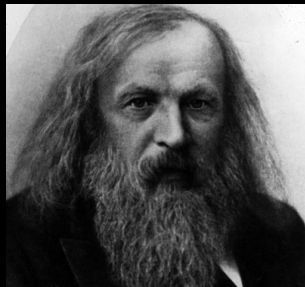




# Example 1.

## The Periodic Table

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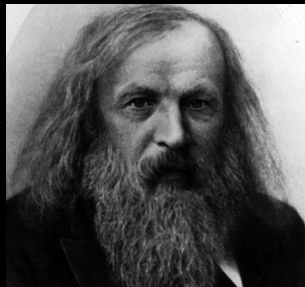


|    | Gruppe I.<br>—<br>H <sup>0</sup> | Gruppe II.<br>—<br>H <sup>0</sup> | Gruppe III.<br>—<br>R <sup>0</sup> | Gruppe IV.<br>R <sup>1</sup><br>R <sup>0</sup> | Gruppe V.<br>R <sup>1</sup><br>R <sup>0</sup> | Gruppe VI.<br>R <sup>1</sup><br>R <sup>0</sup> | Gruppe VII.<br>R <sup>1</sup><br>R <sup>0</sup> | Gruppe VIII.<br>—<br>R <sup>0</sup> |
|----|----------------------------------|-----------------------------------|------------------------------------|------------------------------------------------|-----------------------------------------------|------------------------------------------------|-------------------------------------------------|-------------------------------------|
| 1  | H=1                              |                                   |                                    |                                                |                                               |                                                |                                                 |                                     |
| 2  | Li=7                             | Be=9,4                            | B=11                               | C=12                                           | N=14                                          | O=16                                           | F=19                                            |                                     |
| 3  | Na=23                            | Mg=24                             | Al=27,5                            | Si=28                                          | P=31                                          | S=32                                           | Cl=35,5                                         |                                     |
| 4  | K=39                             | Ca=40                             | —=44                               | Ti=48                                          | V=51                                          | Cr=52                                          | Mn=55                                           | Fe=56, Co=59, Ni=59, Cu=63.         |
| 5  | (Ca=63)                          | Zn=65                             | —=68                               | —=72                                           | As=75                                         | Se=78                                          | Br=80                                           |                                     |
| 6  | Rb=85                            | Str=87                            | Yt=88                              | Zr=90                                          | Nb=94                                         | Mo=96                                          | —=100                                           | Ra=104, Rn=104, Po=106, Ag=108.     |
| 7  | (Ag=108)                         | Cd=112                            | In=115                             | Su=118                                         | Sh=122                                        | Tu=125                                         | J=127                                           |                                     |
| 8  | Cs=133                           | Ba=137                            | Pd=138                             | Ce=140                                         | —                                             | —                                              | —                                               |                                     |
| 9  | (—)                              | —                                 | —                                  | —                                              | —                                             | —                                              | —                                               |                                     |
| 10 | —                                | —                                 | Er=178                             | La=180                                         | Ta=182                                        | W=184                                          | —                                               | Os=195, Ir=197, Pt=198, Au=199.     |
| 11 | (Au=199)                         | Hg=200                            | Tl=204                             | Pb=207                                         | Bi=208                                        | —                                              | —                                               |                                     |
| 12 | —                                | —                                 | —                                  | Th=231                                         | —                                             | U=240                                          | —                                               |                                     |

# Example 1.

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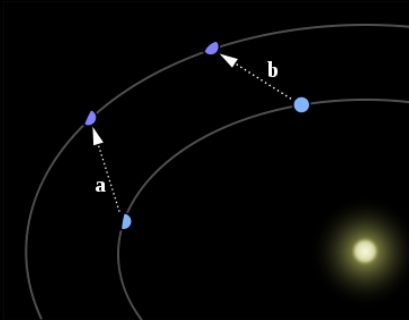
- ▶ In 1865, Mendeleev proposed that elements be ordered into the periodic table.
- ▶ He left gaps in the table, where his theory predicted there should be new matter.
- ▶ When these elements were discovered, he was hailed as a scientific hero.



| Atom<br>Nr. | Gruppe I.<br>—<br>H <sup>0</sup> | Gruppe II.<br>—<br>H <sup>0</sup> | Gruppe III.<br>—<br>R <sup>0</sup> | Gruppe IV.<br>—<br>R <sup>0</sup> | Gruppe V.<br>—<br>R <sup>0</sup> | Gruppe VI.<br>—<br>R <sup>0</sup> | Gruppe VII.<br>—<br>R <sup>0</sup> | Gruppe VIII.<br>—<br>R <sup>0</sup> |
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## Example 2.

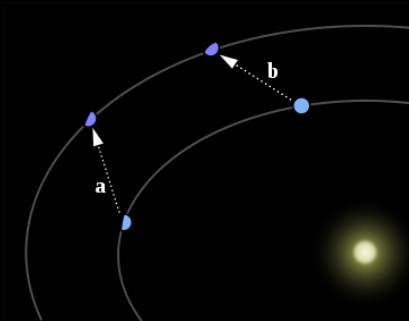
### Orbits of Uranus and Mercury



- In 1841, it was noticed that Uranus' orbit was slightly off from predictions.

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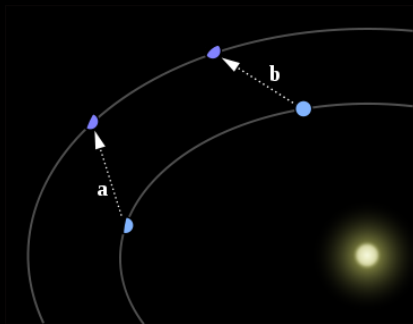
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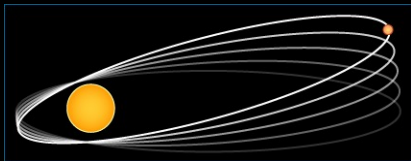
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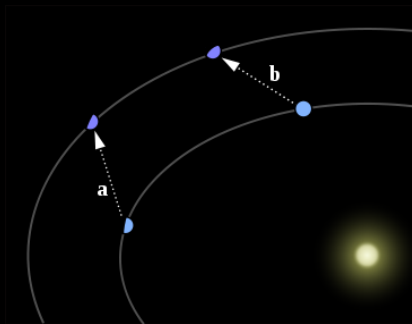
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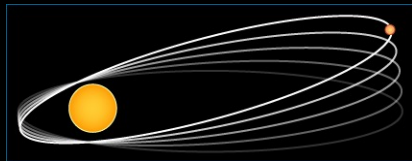
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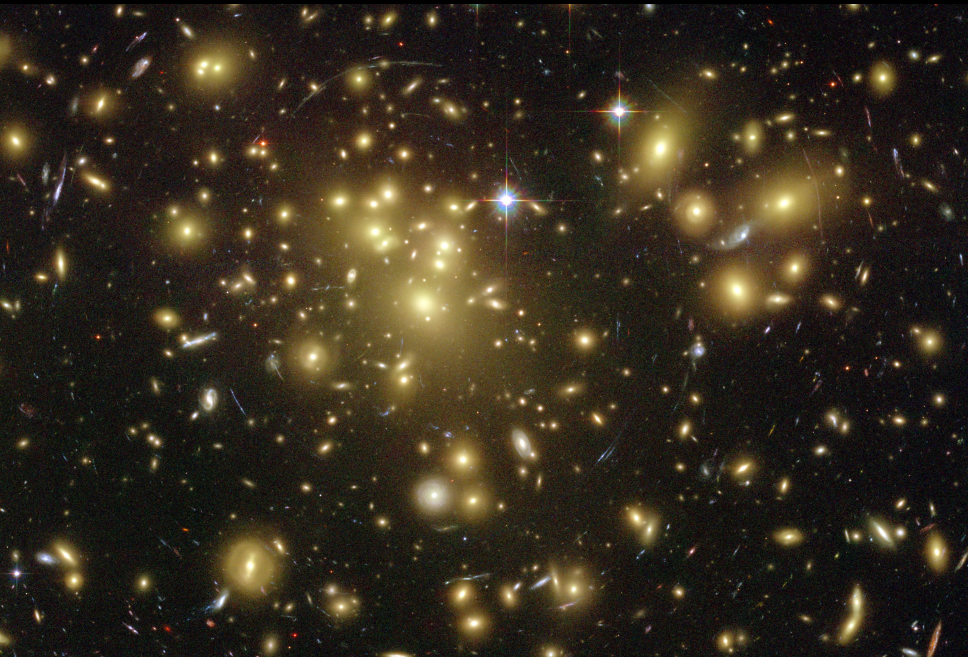
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- ▶ In 1859, a similar problem emerged with the orbit of mercury
- ▶ This issue wasn't solved until Einstein's theory of general relativity superseded Newton's gravity in 1915.

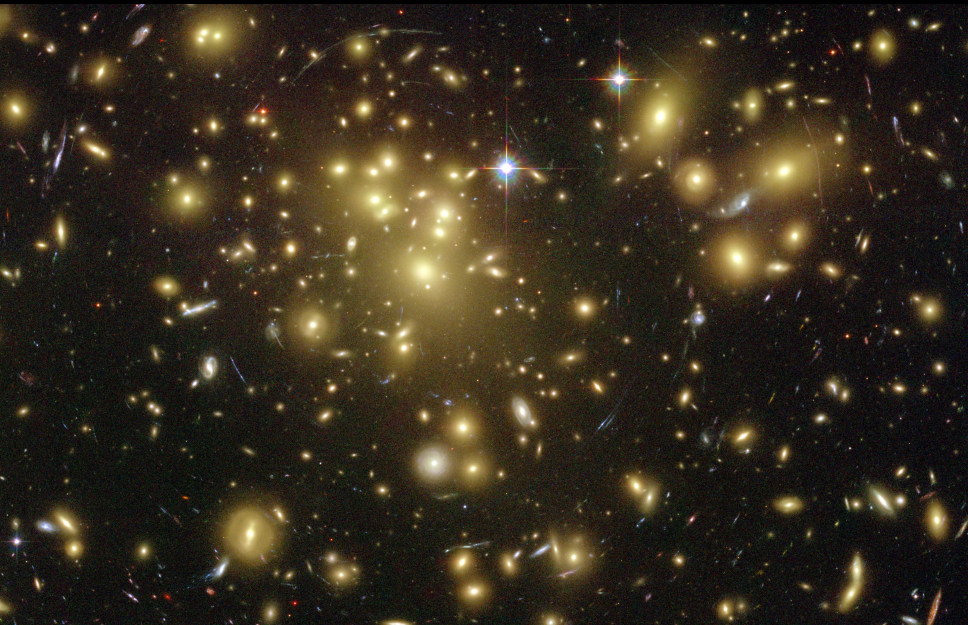


## Other evidence

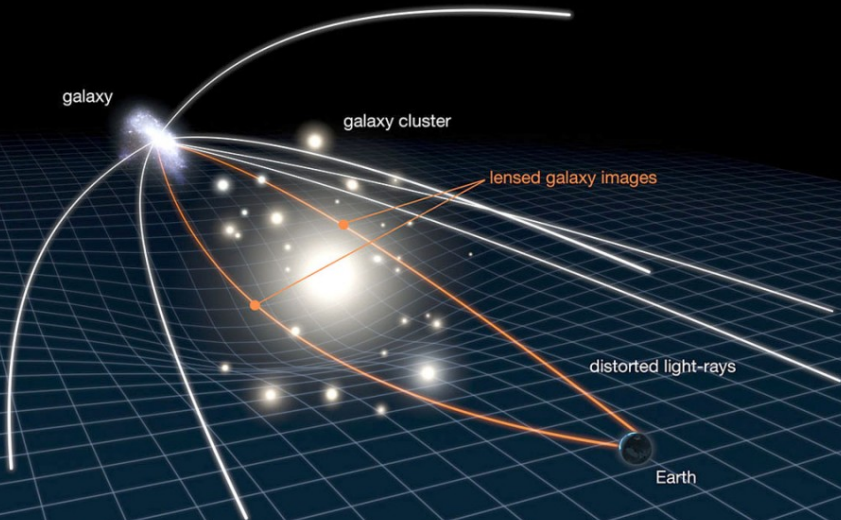


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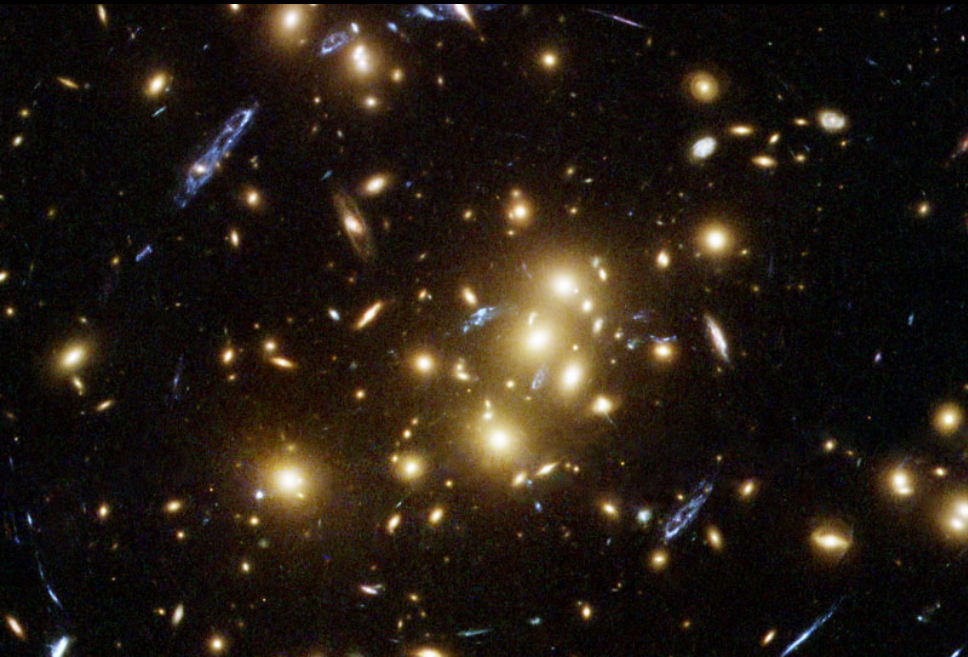
## Gravitational Lensing



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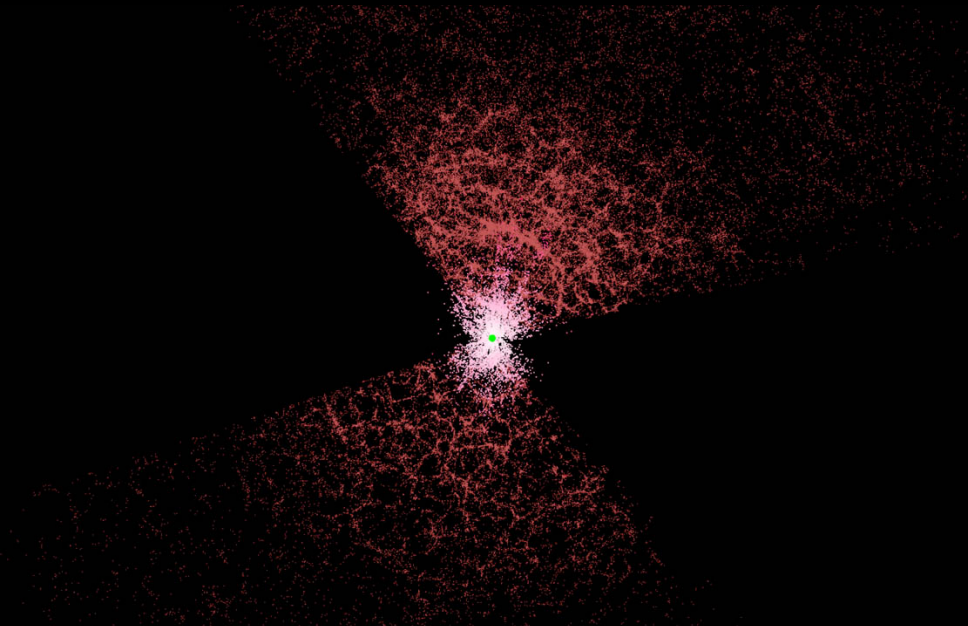
Structure formation





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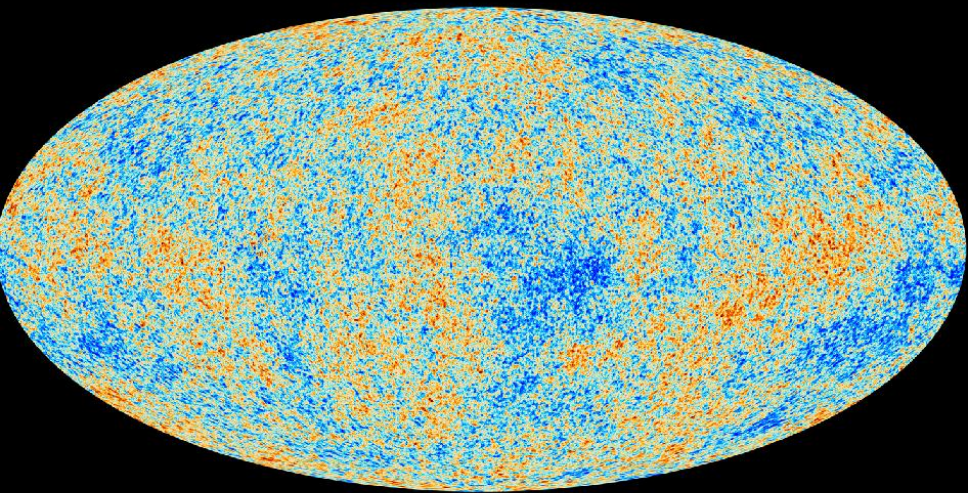
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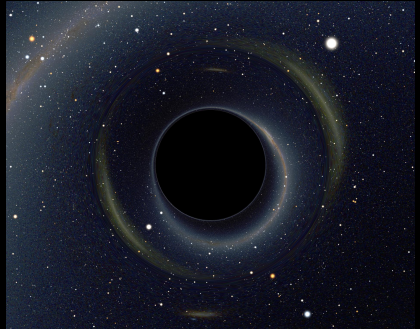
## Other evidence

Cosmic Microwave Background



# What could dark matter be?

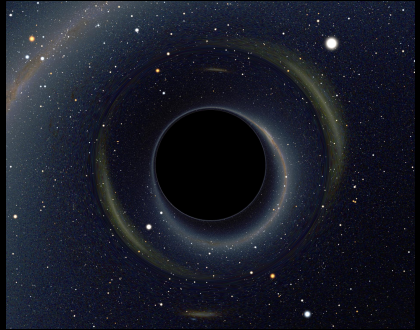
MACHOs: Massive  
Astrophysical  
Compact Halo  
Objects.



# What could dark matter be?

**MACHOs:** Massive  
Astrophysical  
Compact Halo  
Objects.

**WIMPs:** Weakly Interacting  
Massive Particles.



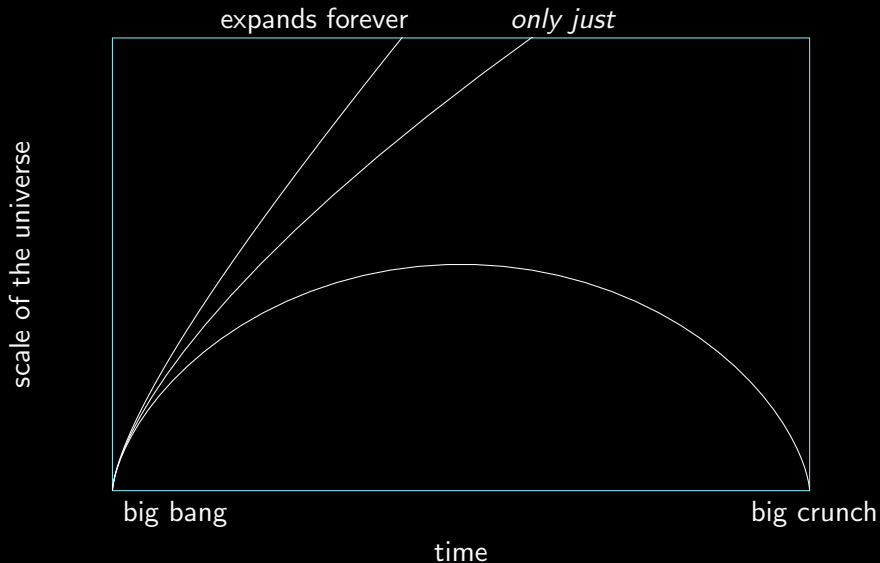
# Dark Energy

## The Problem



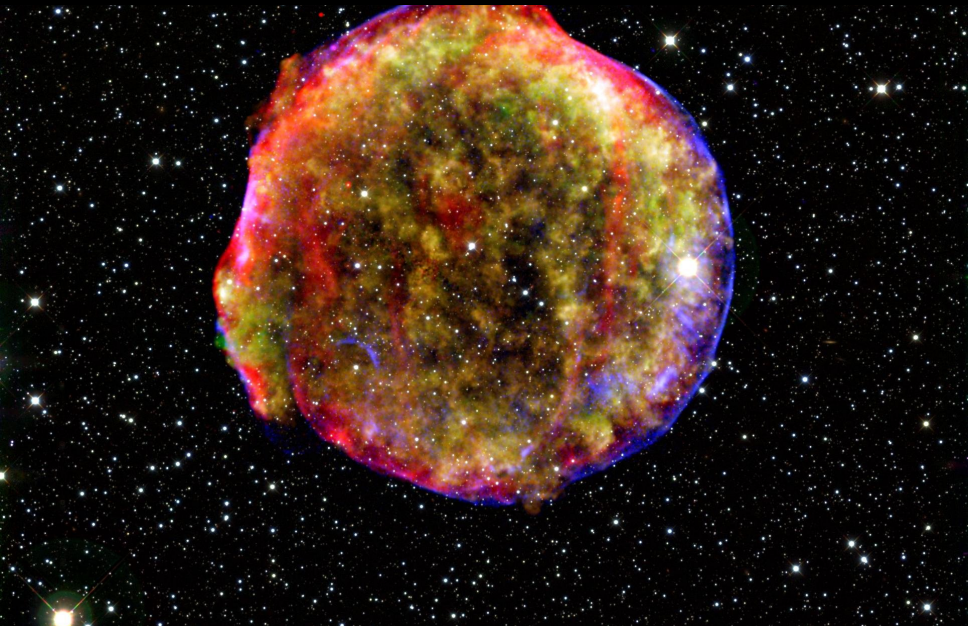
# The evolution of the universe

Three options:



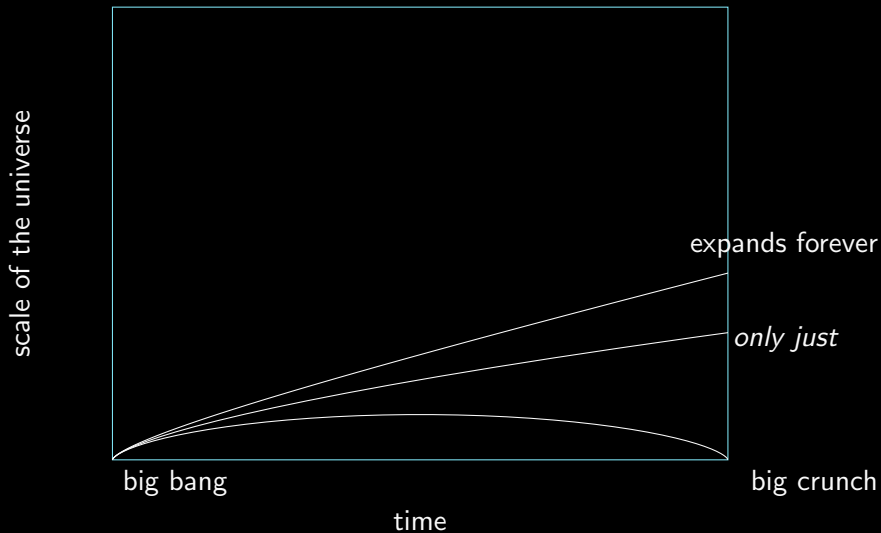
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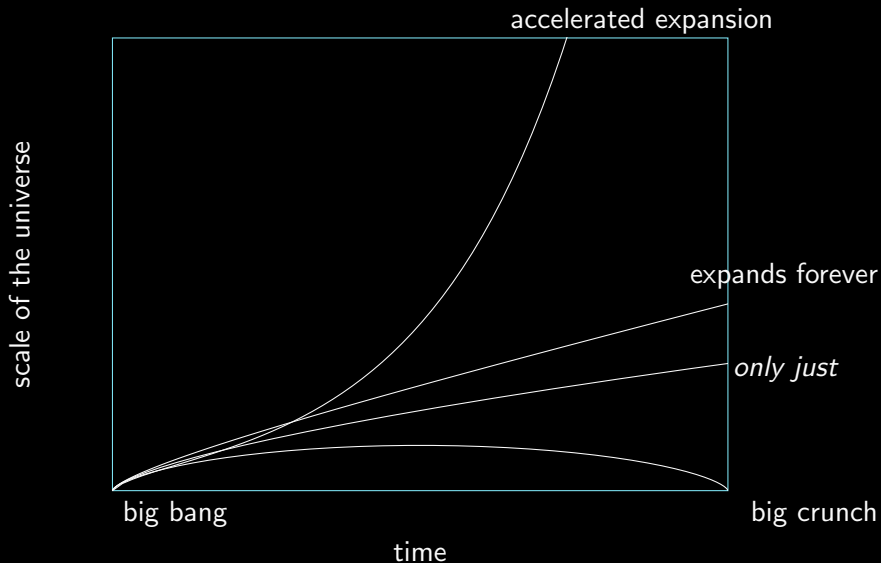
Three options:





# The evolution of the universe

None of the above:



# Dark Energy

## The Problem

- ▶ The Universe isn't slowing down in its expansion, but *speeding up!*

# Dark Energy

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# Dark Energy

## The Problem

- ▶ The Universe isn't slowing down in its expansion, but *speeding up!*
- ▶ One of the ways in which this could happen would be the presence of an energy permeating the universe, accelerating the expansion.
- ▶ We call this Dark Energy
- ▶ Like Dark Matter, it's really just a name for a phenomenon which we observe, and haven't been able to explain yet

# Dark Energy...

...or Cosmological Constant?

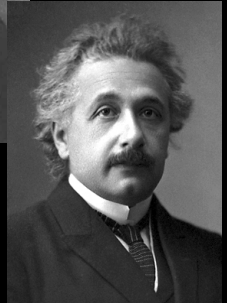
- ▶ Einstein's gravity can be repulsive at large scales.



# Dark Energy...

...or Cosmological Constant?

- ▶ Einstein's gravity can be repulsive at large scales.
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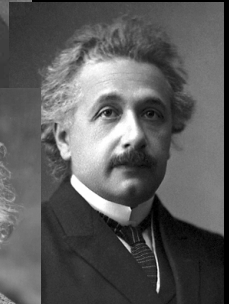
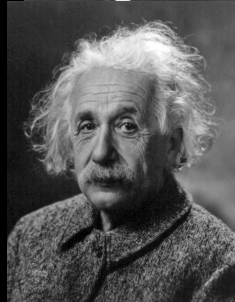




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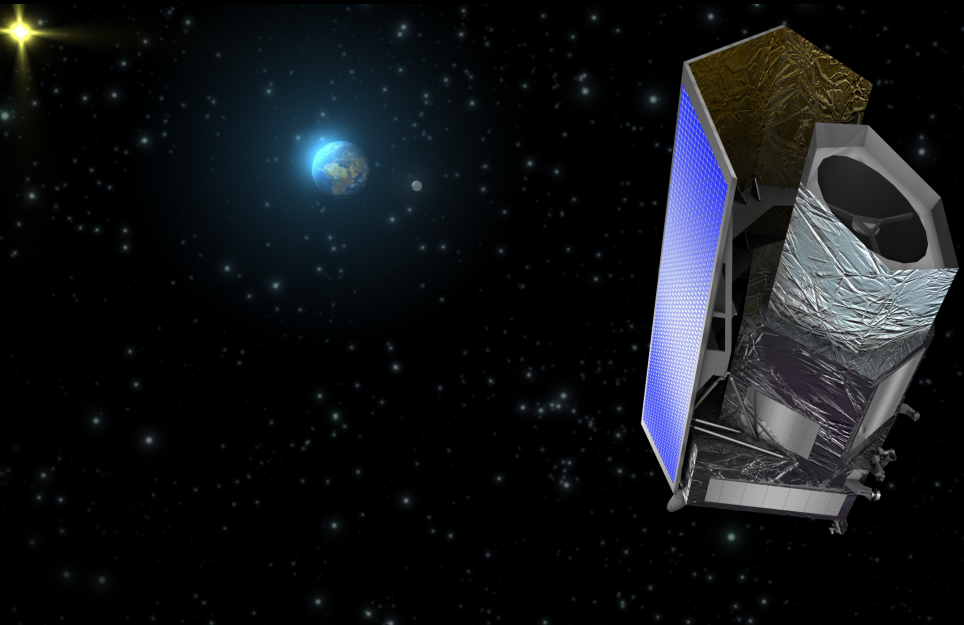
...or Cosmological Constant?

- ▶ Einstein's gravity can be repulsive at large scales.
- ▶ This is known as a cosmological constant.
- ▶ In modern terms, we think of it as a type of energy.



# The Euclid Satellite

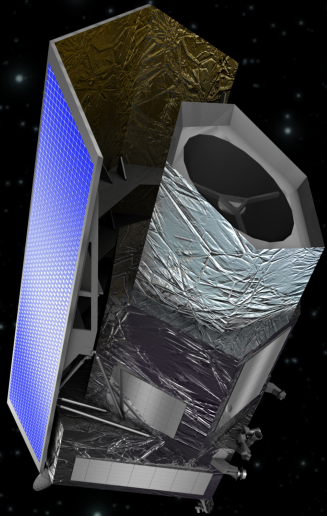
Exploring dark energy



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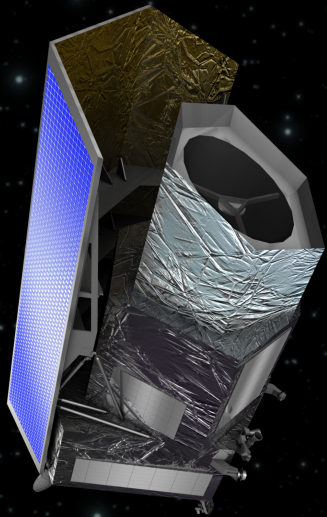
► Launch date  $\approx$  2020



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Exploring dark energy

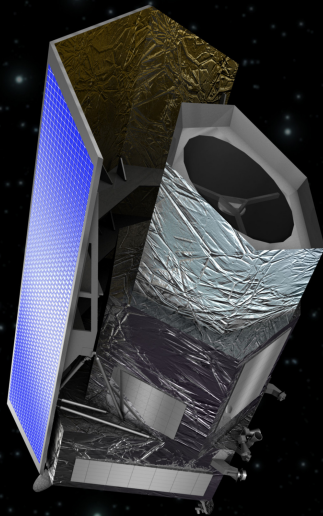
- ▶ Launch date  $\approx$  2020
- ▶ Build up an accurate 4D picture of the visible universe



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Exploring dark energy

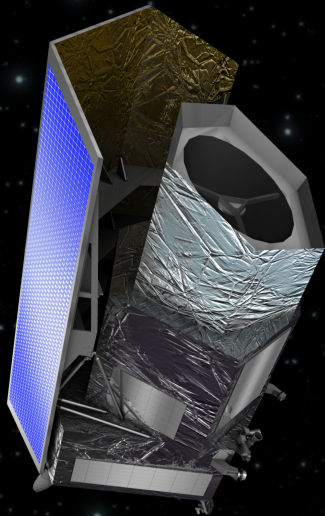
- ▶ Launch date  $\approx$  2020
- ▶ Build up an accurate 4D picture of the visible universe
- ▶ By measuring cosmic shear around galaxies and clusters, will be able to build up a map of Dark Matter



# The Euclid Satellite

Exploring dark energy

- ▶ Launch date  $\approx$  2020
- ▶ Build up an accurate 4D picture of the visible universe
- ▶ By measuring cosmic shear around galaxies and clusters, will be able to build up a map of Dark Matter
- ▶ Will see any variation in dark energy across space and time, enabling us to build a better understanding of it.



# The Composition of the Universe

